

Professional Summary

AI/ML Engineer with 3+ years of experience designing, deploying, and scaling Generative AI and NLP solutions across healthcare and enterprise domains. Proven expertise in building production-ready LLM-powered applications, including RAG systems, clinical automation tools, ICD-10 prediction models, and secure PHI anonymization pipelines. Skilled in fine-tuning open-source LLMs (LLaMA, BERT, Codex-style models) with LoRA/PEFT, integrating EHR and enterprise APIs, and orchestrating LLMOps pipelines with Azure ML, MLflow, and drift detection frameworks. Adept at leveraging vector databases (Pinecone, FAISS, ChromaDB) and LangChain to develop retrieval-based systems, chatbots, and summarization engines that improve efficiency and compliance. Strong background in HIPAA-compliant AI deployments, explainability dashboards (Power BI), and cross-functional collaboration with compliance, clinical, and IT teams. Recognized for mentoring engineers in LLM fine-tuning, prompt engineering, and privacy-first AI design, driving organizational adoption of GenAI technologies.

Technical Skills

- **Programming & Scripting:** Python, SQL, FastAPI, Streamlit, Docker, Kubernetes
 - **LLMs & GenAI:** GPT-3.5/4, LLaMA 2, Codex, OpenAI API, Hugging Face Transformers, LoRA, PEFT
 - **Retrieval & Search (RAG):** LangChain, Pinecone, ChromaDB, FAISS, Vector Databases
 - **Machine Learning & NLP:** BERT, NER Tagging, ICD-10 Classification, Text Summarization, Embeddings
 - **MLOps & Deployment:** Azure ML, MLflow, Apache Airflow, CI/CD, Model Drift Detection
 - **Data & Privacy:** HIPAA Compliance, PHI Anonymization, Explainability Tools, Token Usage Tracking
 - **Monitoring & Analytics:** Power BI, Streamlit Dashboards, BLEU Score Evaluation, Model Explainability
 - **Collaboration Tools:** Jira, Confluence, EHR API Integration, RFP Automation
 - **Domains:** Healthcare AI, Clinical NLP, DevOps Automation, Recruitment AI
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Professional Experience

GE Healthcare | USA

AI/ML Engineer | Aug 2024 – Present

- Designed and deployed a **GPT-4-powered claim validation engine** using OpenAI API and embeddings, reducing manual review effort by **65%**.
 - Built a **secure RAG-based patient query bot** with Pinecone, LangChain, and custom retrievers, improving internal clinical query resolution speed by **3x**.
 - Automated **medical summary generation** from physician notes using LangChain, FAISS, and multi-step LLM chaining, reducing review time by clinicians.
 - Developed a **HIPAA-compliant PHI anonymization pipeline** using Hugging Face Transformers and NER tagging, ensuring full compliance across AI workflows.
 - Fine-tuned **LLaMA 2 with LoRA adapters** to classify and triage patient emails into actionable categories, increasing operational efficiency.
 - Trained and deployed a **BERT-based ICD-10 code prediction model**, boosting classification accuracy by **40%** over baseline models.
 - Built an **internal chatbot for clinical protocols** using LangChain, OpenAI models, and Streamlit, making compliance and protocol data instantly accessible.
 - Integrated **EHR APIs with GenAI pipelines** via Prompt Templates and automation scripts, enabling seamless retrieval and structured responses.
 - Established **LLMOps pipelines** with Azure ML, MLflow, and drift detection tooling, tracking token usage, latency, and evaluation metrics in production.
 - Created **Power BI dashboards for explainability** and compliance reporting, providing real-time monitoring of model fairness, privacy, and audit readiness.
 - Collaborated with cross-functional teams (compliance, clinical, IT) to ensure **trustworthy, transparent, and secure AI deployments**.
 - Mentored internal teams on **best practices for LLM fine-tuning, prompt engineering, and privacy-first AI design**, upskilling engineers in generative AI adoption.
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Evolvemed | Delhi, IND

AI/ML Engineer | Jan 2022 – July 2023

- Built a **GenAI-powered DevOps assistant** with GPT-3.5, LangChain, and FastAPI to recommend CI/CD pipeline fixes, reducing engineering downtime.
- Automated **Jira and Confluence summarization** using OpenAI embeddings, cutting manual documentation review time by **55%**.
- Developed **vector-search knowledge bases** using Pinecone, ChromaDB, and LangChain, enabling engineers to retrieve technical documentation in seconds.
- Fine-tuned **Codex-style LLMs with PEFT** to auto-generate inline code comments and documentation, improving developer productivity and reducing errors.
- Implemented **AI-driven resume filtering** using embeddings, cosine similarity, and FastAPI, streamlining hiring pipelines and improving recruiter efficiency.
- Deployed **RFP summarization pipelines** with Hugging Face Transformers to assist project managers in evaluating large proposals quickly.
- Built **internal query bots** with LangChain, Prompt Templates, and Streamlit, improving support and knowledge sharing for engineering teams.
- Managed **end-to-end GenAI orchestration** with Apache Airflow, Docker, and Kubernetes, ensuring reliable and scalable deployment pipelines.
- Set up **LLM evaluation metrics (BLEU scores, embedding similarity)** in MLflow with dashboards for monitoring quality and drift.
- Created a **bug triaging chatbot** using GPT-3.5 and vectorized historical QA logs, significantly improving bug assignment efficiency.
- Designed **internal RAG workflows for developer knowledge bases**, reducing onboarding time for new engineers.
- Worked cross-functionally with QA, product, and DevOps teams to **embed GenAI in developer tooling**, driving adoption across engineering processes.

Education

Masters in Computer science - **New Jersey Institute of Technology**

CERTIFICATIONS

- Academy Accreditation - Generative AI Fundamentals
 - Bloomberg Market Concepts
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PROJECTS

Project 1 Heart Attack Risk Prediction Web App

- Python, Scikit-learn, Streamlit, Pandas, NumPy, Matplotlib, Streamlit Cloud
- Developed a Streamlit-based ML web app to predict heart attack risk using health metrics.
- Trained KNN, SVM, Random Forest, Logistic Regression, and Decision Tree; achieved 98.4% accuracy.
- Performed EDA, feature selection, and deployed on Streamlit Cloud for scalability.

Project 2 Tomato Plant Disease Detection Using CNN

- Python, CNN, TensorFlow, Keras, Image Processing
- Built a CNN model to classify tomato plant diseases (Early Blight, Late Blight, Bacterial Spot) from leaf images.
- Achieved 98.74% accuracy, enabling early disease detection and supporting precision agriculture.
- Promotes sustainable farming with optimized resource usage and reduced crop loss.

Project 3 Mercedes-Benz Stock Price Prediction (LSTM)

- Python, LSTM, Keras, Pandas, Matplotlib, Jupyter
 - Built an LSTM-based deep learning model to forecast Mercedes-Benz stock prices using historical time-series data.
 - Evaluated using MAE, MSE, and RMSE; visualized prediction trends and confidence intervals.
 - Hosted project repository with data, notebooks, trained models, and output visualizations.
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EDUCATION

Masters in Computer Science – **New Jersey Institute of Technology (NJIT)**

Sep 2023 –

Dec 2024

Bachelors in Mechanical Engineering – **ACE Engineering College**

Jun 2019 – Jul 2023

Github Link: <https://github.com/Rahhuul07>

Portfolio: <https://rahhuul07.github.io/My-Portfolio/>